

Simultaneous partition of hypergraphs

OPG-46817

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Problem Let H_1 and H_2 be two r -uniform hypergraph on the same vertex set V . Does there always exist a partition of V into r classes $V_1, \dots, V_r$ such that for both $i = 1, 2$, at least $r!m_i/r^r - o(m_i)$ hyperedges of H_i meet each of the classes $V_1, \dots, V_r$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>